

Week 4. Problem set (solutions by Karim Khabibrakhmanov)

1. Compute asymptotic worst case time complexity of `solve` in terms of the size n of the input list:

- (i) Express the running time of the `solve` procedure $T(n)$ as a recurrence relation.
- (ii) Find the asymptotic complexity of $T(n)$ using the master theorem.
- (iii) Specify which case of the master theorem applies (if any).
- (iv) Write down conditions for this case that need to be checked (write down specialized version for a particular recurrence).
- (v) Prove that the conditions are satisfied. For asymptotic notation, each property used in a proof **must** be either proven explicitly or properly referenced (e.g. by citing [CLRS] or a particular Lecture/Tutorial slide).

```

1  /* L is a linked list */
2  solve(L):
3      if L is empty
4          return 1
5      else
6          s = 1
7          B = new linked list
8          for j from 1 to 3 /* inclusive */
9              i = 0
10             A = L.head
11             while A.next is not null
12                 b = (i >> j) & 3 /* jth and (j+1)th least significant bits of i */
13                 if b == 1
14                     B.add(i / 2, A.value)
15                 A = A.next
16                 i = i + 1
17             s = s + helper(B)

```

Answer:

- (i) $T(n) = 3 \cdot T(\frac{n}{4}) + \Theta(n)$
- (ii) $T(n) = \Theta(n)$
- (iii) Case 3.
- (iv) If there exists a constant $\epsilon > 0$ such that $f(n) = \Omega(n^{\log_b a + \epsilon})$, and if $f(n)$ additionally satisfies the regularity condition $a \cdot f(\frac{n}{b}) \leq c \cdot f(n)$ for some constant $c < 1$ and all sufficiently large n , then $T(n) = \Theta(f(n))$.
- (v) $3 \cdot \frac{n}{4} \leq c \cdot n$, if $c = \frac{3}{4} < 1 \rightarrow n \leq n$

Justification:

The loop iterates three times, matching the number of recursive calls.

The expression $b = (i \gg j) \& 3$ adds every fourth element to list B because the two bits extracted by $(i \gg j) \& 3$ repeat cyclically every four numbers, and only the combination 01 ($b == 1$) satisfies the condition. The function $f(n)$ is equal to $\Theta(n)$ because the reduction in the size of the list runs in linear time.

For $T(n) = 3 \cdot T(\frac{n}{4}) + \Theta(n)$, where $a = 3 > 0$, $b = 4 > 1$, and $f(n) = \Theta(n)$, the recurrence relation satisfies the conditions of the Master Theorem:

- $\log_b a = \log_4 3 \approx 0.793$
- Case 1:
 $f(n) = \Theta(n) = O(n^{\log_b a - \epsilon}) = O(n^{0.793 - \epsilon})$ - No, because $\epsilon > 0$ and $\Theta(n)$ grows faster.

- Case 2:
 $f(n) = \Theta(n) = \Theta(n^{\log_b a} \cdot \log^k n) = \Theta(n^{0.793} \cdot \log^k n)$ - No, because $\Theta(n)$ grows faster than $\Theta(n^{0.793} \cdot \log^k n)$ for any k .
- Case 3:
 $f(n) = \Theta(n) = \Omega(n^{\log_b a + \epsilon}) = \Omega(n^{0.793 + \epsilon})$ - Yes. For $\epsilon \approx 0.2$, $n^{0.793 + \epsilon}$ satisfies the condition.
 [[CLRS], Theorem 4.1 (Master theorem)] Let's check the regularity condition $a \cdot f\left(\frac{n}{b}\right) \leq c \cdot f(n)$:
 $3 \cdot \Theta\left(\frac{n}{4}\right) \leq c \cdot \Theta(n)$
 $3 \cdot \frac{n}{4} \leq c \cdot n$, if $c = \frac{3}{4} < 1$
 $n \leq n$. This condition is satisfied.

Answer: $\Theta(n)$

2. Consider the following modification of the binary search algorithm [CLRS, Exercise 2.3-6]: given an input array A of size n , a parameter k , and an integer value x
- if the array is empty — stop with failure (x not found)
 - if the size of the input array is 1 and $A[0] = x$ — stop with success (x is found)
 - otherwise, we create midpoints m_i (for all integer indices $1 \leq i \leq k$) *evenly* spaced between start index $m_0 = 0$ and end index $m_{k+1} = n - 1$
 - we then find consecutive indices m_i and m_{i+1} such that $A[m_i] \leq x \leq A[m_{i+1}]$ and repeat the search on the subarray $A[m_i : m_{i+1}]$

Compute the time complexity of this algorithm in terms of n and k using two different methods:

- Write down the running time $T(n)$ as a recurrent formula.
- Apply the master theorem and get closed form formula for $T(n)$ (assume k is a constant, the final answer may be independent of k):
 - Which case of the master theorem applies?
 - Write down the conditions for this case that need to be checked (write down specialized version for this particular recurrence).
 - Provide asymptotic complexity for $T(n)$ using Θ -notation.
- Apply the substitution method and get closed form formula for $T(n, k)$ (treat k as a parameter, the final answer should depend on k).
- Which value of k is optimal? Justify your answer (4–6 sentences).

Answer:

- $T(n) = T(\frac{n}{k+1}) + \Theta(k)$
- $T(n) = \Theta(\log n)$
- $T(n, k) = \Theta(k \cdot \log_{k+1} n)$
- Optimal value of k equal to 1

Justification:

- The general form of the recurrence relation is as follows:

$$T(n) = a \cdot T\left(\frac{n}{b}\right) + f(n)$$

In the binary search algorithm:

- We have $a = 1$, because the function is called only once to repeat the search on the subarray $A[m_i : m_{i+1}]$.
- $b = k + 1$, because the algorithm divides the array into $k + 1$ subarrays.

Thus, the recurrence relation for the algorithm is:

$$T(n) = T\left(\frac{n}{k+1}\right) + \Theta(k)$$

- For $T(n) = T(\frac{n}{k+1}) + \Theta(k)$, where $a = 1 > 0$, $b = k + 1 > 1$ ($k \geq 1$), and $f(n) = \Theta(k)$, the recurrence relation satisfies the conditions of the Master Theorem:

- $\log_b a = \log_{k+1} 1 = 0$
- Case 1:
 $f(n) = \Theta(k) \neq O(n^{-\epsilon})$ for any $\epsilon > 0$, because $\Theta(k)$ is bounded below by a positive constant, while $O(n^{-\epsilon})$ tends to zero as $n \rightarrow \infty$.
- Case 2:
 $f(n) = \Theta(k) = \Theta(n^{\log_b a} \cdot \lg^k n) = \Theta(n^0 \cdot \lg^k n) = \Theta(\lg^k n)$ - Yes, if $k = 0$.

Therefore, $\Theta(\log^1 n) = \Theta(\log n)$

(iii) The substitution method:

Suppose $n = b^i = (k+1)^i$

$$T((k+1)^i) = T((k+1)^{i-1}) + \Theta(k)$$

$$T((k+1)^i) = T((k+1)^{i-2}) + \Theta(k) \cdot 2$$

$$T((k+1)^i) = T((k+1)^{i-3}) + \Theta(k) \cdot 3$$

$\dots$

$$T((k+1)^i) = T(1) + \Theta(k) \cdot i$$

$$i = \log_{k+1} n, T(n) = T(1) + \Theta(k) \cdot \log_{k+1} n$$

$$T(n) = \Theta(k \cdot \log_{k+1} n)$$

(iv) Optimal value of k is 1 because $k \geq 1$ and the base of the logarithm satisfies the condition since $k+1 \geq 2 > 1$. Since $\Theta(k \cdot \log_{k+1} n)$ increases with larger values of k , it is advantageous to choose the smallest possible value.

3. For each of the following recurrences, apply the master theorem [CLRS, Theorem 4.1] yielding a closed form formula for the asymptotic complexity of $T(n)$. Assume that $T(n) = 1$ when $n < 10$ for all recurrences below. You must specify the following:

- Can the theorem be applied to the recurrence?
- If yes, then
 - (i) Which case of the master theorem applies?
 - (ii) Which conditions for this case that need to be checked (write down specialized version for a particular recurrence).
 - (iii) Prove that the condition is satisfied. For asymptotic notation, each property used in a proof **must** be either proven explicitly or properly referenced (e.g. by citing [CLRS] or a particular Lecture/Tutorial slide).
 - (iv) Provide asymptotic complexity for $T(n)$ using Θ -notation.
- Otherwise, provide an explicit justification, explaining why the theorem cannot be applied.

(a) $T(n) = 6T(n/36) + \sqrt{n} \cdot \log_2^2 n$

(b) $T(n) = 4T(3n/8) + n^2$

(c) $T(n) = 7T(n/5) + \log_2(n!)$

(d) $T(n) = 3T(\sqrt[3]{n}) + n$

(e) $T(n) = \frac{1}{3}T(3n) + n \log_3 n$

Answer:

(a) $T(n) = \Theta(\sqrt{n} \cdot \lg^3 n)$

(b) $T(n) = \Theta(n^2)$

(c) $T(n) = \Theta(n^{\log_5 7})$

(d) No answer

(e) No answer

Justification:

(a) For $T(n) = 6T(n/36) + \sqrt{n} \cdot \log_2^2 n$, where $a = 6 > 0$, $b = 36 > 1$, and $f(n) = \sqrt{n} \cdot \log_2^2 n$, the recurrence relation satisfies the conditions of the Master Theorem:

- $\log_b a = \log_{36} 6 = \frac{1}{2}$
- Case 1:
 $f(n) = \sqrt{n} \cdot \log_2^2 n = O(n^{\log_b a - \epsilon}) = O(n^{\frac{1}{2} - \epsilon})$ - No, because $\epsilon > 0$ and $\sqrt{n} \cdot \log_2^2 n$ grows faster.
- Case 2:
 $f(n) = \sqrt{n} \cdot \log_2^2 n = \Theta(n^{\log_b a} \cdot \lg^k n) = \Theta(n^{\frac{1}{2}} \cdot \lg^k n)$ - Yes, if $k = 2$.

Answer: $\Theta(\sqrt{n} \cdot \lg^{k+1} n) = \Theta(\sqrt{n} \cdot \lg^3 n)$

(b) For $T(n) = 4T(\frac{3n}{8}) + n^2$, where $a = 4 > 0$, $b = \frac{8}{3} > 1$, and $f(n) = n^2$, the recurrence relation satisfies the conditions of the Master Theorem:

- $\log_b a = \log_{\frac{8}{3}} 4 \approx 1.413$
- Case 1:
 $f(n) = n^2 = O(n^{\log_b a - \epsilon}) = O(n^{1.413 - \epsilon})$ - No, because $\epsilon > 0$ and n^2 grows faster.
- Case 2:
 $f(n) = n^2 = \Theta(n^{\log_b a} \cdot \log^k n) = \Theta(n^{1.413} \cdot \log^k n)$ - No, because n^2 grows faster than $n^{1.413} \cdot \log^k n$ for all k .

- Case 3:
 $f(n) = n^2 = \Omega(n^{\log_b a + \epsilon}) = \Omega(n^{1.413 + \epsilon})$ - Yes. For $\epsilon \approx 0.6$, $n^{1.413 + \epsilon}$ satisfies the condition.
 Let's check the regularity condition $a \cdot f\left(\frac{n}{b}\right) \leq c \cdot f(n)$:
 $4 \cdot \frac{9 \cdot n^2}{64} \leq c \cdot n^2$
 $\frac{9 \cdot n^2}{16} \leq c \cdot n^2$, if $c = \frac{9}{16} < 1$
 $n^2 \leq n^2$. This condition is satisfied.

Answer: $\Theta(n^2)$

- (c) For $T(n) = 7T(n/5) + \log_2(n!)$, where $a = 7 > 0$, $b = 5 > 1$, and $f(n) = \log_2(n!)$, the recurrence relation satisfies the conditions of the Master Theorem:

- $\log_b a = \log_5 7 \approx 1.209$
- Case 1:
 $f(n) = \log_2(n!) = O(n^{\log_b a - \epsilon}) = O(n^{1.209 - \epsilon})$ - Yes, because $\log_2(n!)$ does not grow faster than $n^{1.209}$.

Answer: $\Theta(n^{\log_5 7})$

- (d) For $T(n) = 3T(\sqrt[3]{n}) + n$, the recurrence relation does not satisfy the conditions of the Master Theorem, because $T(\sqrt[3]{n})$ does not match the form $T(n/b)$, where $b > 1$ is a constant. Therefore, the Master Theorem cannot be applied directly.
- (e) For $T(n) = \frac{1}{3}T(3n) + n \log_3 n$, the recurrence relation does not satisfy the conditions of the Master Theorem, because $b = \frac{1}{3}$ does not match $b > 1$. Therefore, the Master Theorem cannot be applied directly.

4. (+1% **extra credit**) Consider the following recurrence with parameters $a > 1$ and $b > 1$:

$$T(n) = aT(n/b) + n^{4 \log_a b} \ln^{a-2b} n$$

Specify **precise** conditions on a and b such that

- (i) Case 1 of the master theorem applies to $T(n)$
- (ii) Case 2 of the master theorem applies to $T(n)$
- (iii) Case 3 of the master theorem applies to $T(n)$
- (iv) The master theorem is not applicable to $T(n)$

Provide a brief justification in each case.

Answer: *(replace with your answer)*

Justification: *(replace with your justification)*

References

- [CLRS] Cormen, T.H., Leiserson, C.E., Rivest, R.L. and Stein, C., 2022. *Introduction to algorithms, Fourth Edition*. MIT press.